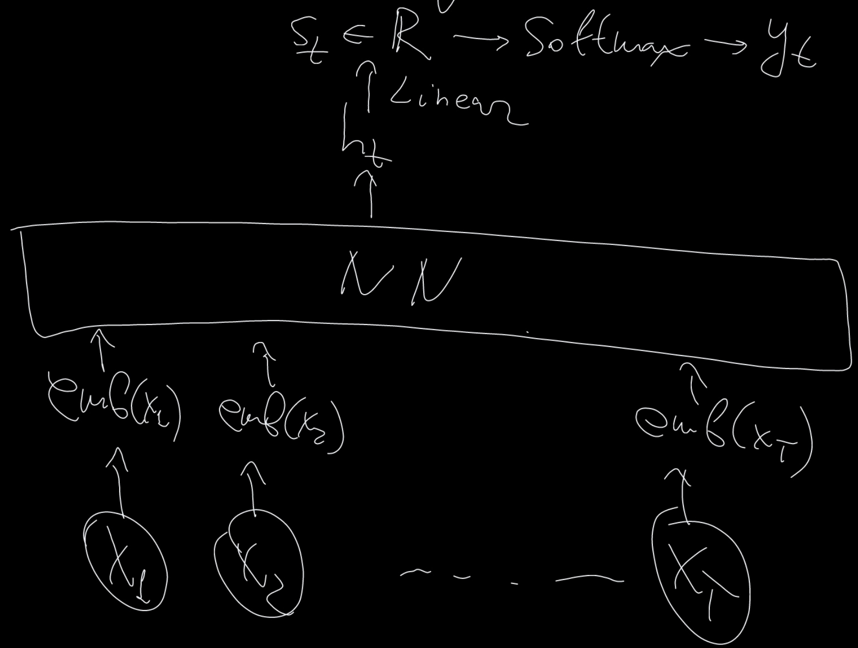
Word Embedding

vocab. size

After tokenizer: $x_t \in \{1, 2, \dots, V\}$



$emb(x_t) \in \mathbb{R}^{emb\_dim}$

Word2Vec

sliding window

text $[\ \boxed{w_1 \quad w_2 \quad w_3 \quad \cdots} \ - - - \quad w_T\ ]$

Variant 1:

$$w_t \to \begin{array}{c} w_{t-M} \\ \vdots \\ w_{t-1} \\ w_{t+1} \\ \vdots \\ w_{t+M} \end{array}$$

Variant 2:

$$\begin{array}{c} w_{t-M} \\ \vdots \\ w_{t-1} \\ w_{t+1} \\ \vdots \\ w_{t+M} \end{array} \to w_t$$

Var 1

$$\text{loss}: -\sum_t \sum_{c \in C_t} \log p(w_c \mid w_t) =$$

$$= -\sum_t \sum_{c \in C_t} \log \frac{\exp(\text{emb}(w_c)^T \text{emb}(w_t))}{\sum_{j=1}^{V} \exp(\text{emb}(w_j)^T \text{emb}(w_t))} \to \min_{\text{emb}}$$

Var 2

$$\text{loss} = -\sum_t \log \frac{\exp(\text{emb}(w_t)^T \sum_{c \in C_t} \text{emb}(w_c))}{\sum_{j=1}^{V} \exp(\text{emb}(w_j)^T \sum_{c \in C_t} \text{emb}(w_c))} \to \min_{\text{emb}}$$

negative sampling:

$$\text{loss} = -\sum_t \left(\sum_{c \in C_t} \log \sigma\left(\text{emb}(w_c)^T \text{emb}(w_t)\right) + \right.$$

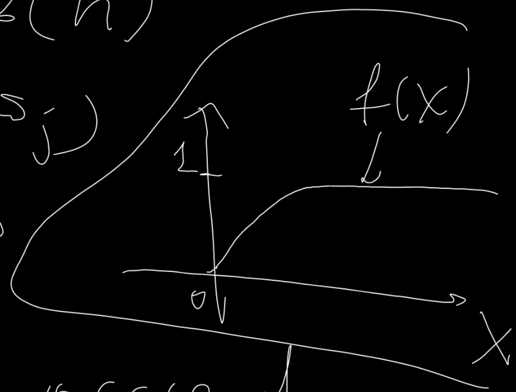
$$\left. + \sum_{n \in N_t} \log \sigma\left(-\text{emb}(w_n)^T \text{emb}(w_t)\right)\right) \to \min_{\text{emb}}$$

fastText

$$\text{score}(w_i, w_j) =$$

$$= \text{emb}(w_i)^T \sum_{n \in n\text{-grams}(w_j)} \text{emb}(n)$$

<where> $\longrightarrow$ <wh, whe, her, ere, re>



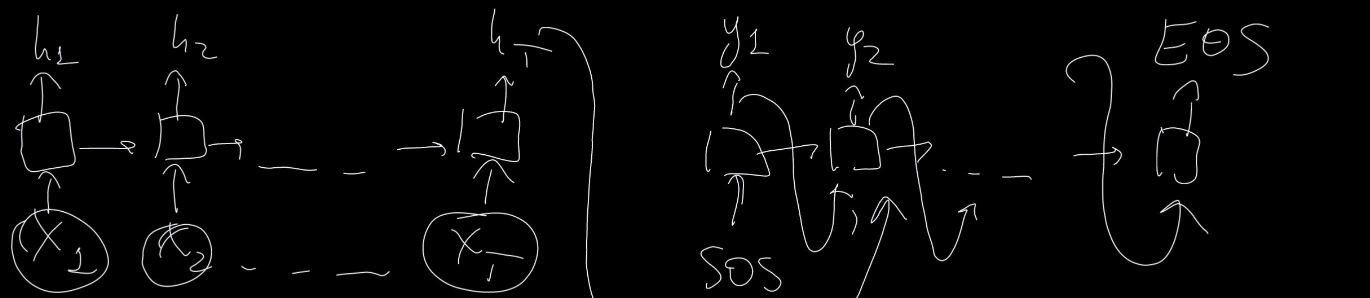
GloVe

X_{ij} – how many times w_i is occured in context of w_j

$$\sum_{i,j} f(X_{ij}) \left(\text{emb}(w_i)^T \text{emb}(w_j) + b_i + b_j - \log X_{ij} \right)^2 \longrightarrow \min_{\text{emb}}$$

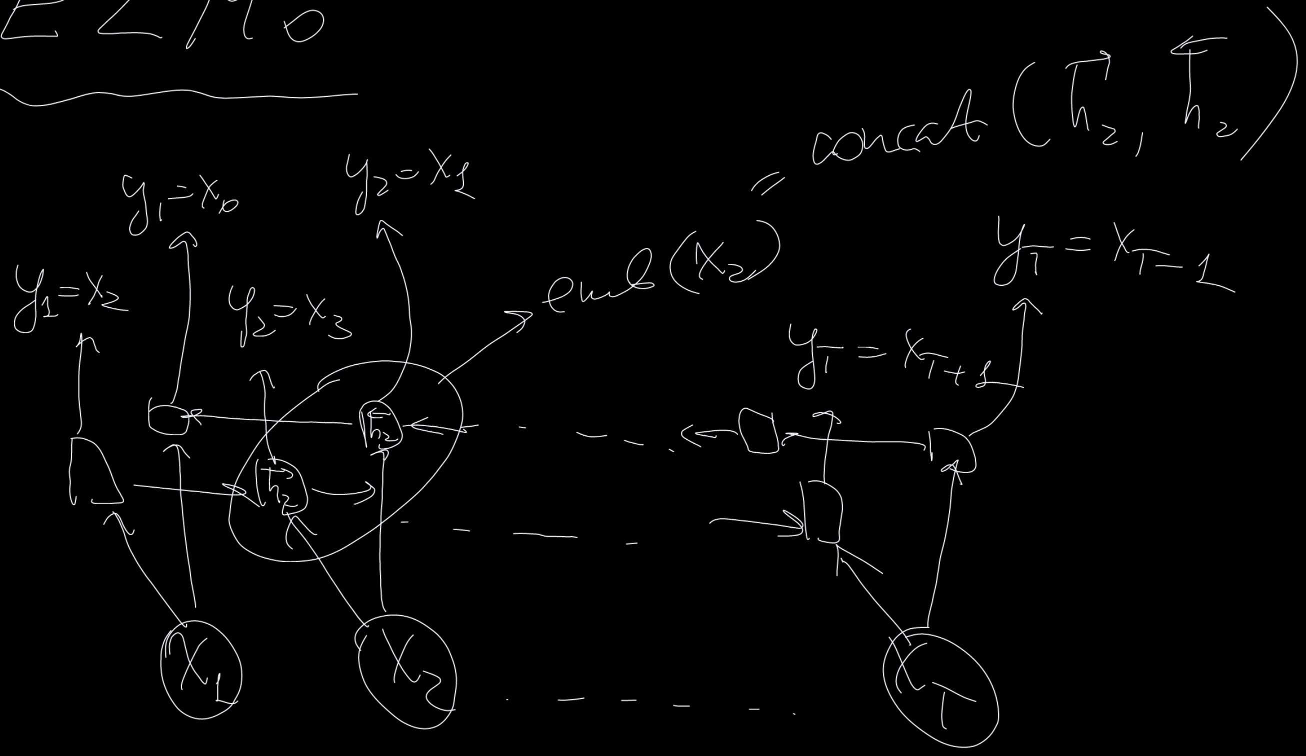
CoVe

Machine translation task



Use only encoder for word embeddings

ELMo



BERT-like
word embedding

Transformer

"query" "keys" "values"

$$\text{output} = \text{Attention}(q, k, v)$$

$$\alpha_i = \frac{\exp(q^T k_i)}{\sum_{j=1}^{I} \exp(q^T k_j)}$$

$$\text{output} = \sum_{j=1}^{I} \alpha_j v_j$$

$$q = h_{k-1}$$

$$k = \{h_1, h_2, \ldots, h_I\}$$

$$v = \{h_1, h_2, \ldots, h_I\}$$

$$\underset{\mathbb{R}^{\tilde{T}\times H}}{\underset{\in}{O}} = \mathrm{Att}\left(\underset{\mathbb{R}^{\tilde{T}\times H}}{\underset{\in}{Q}},\ \underset{\mathbb{R}^{T\times H}}{\underset{\in}{K}},\ \underset{\mathbb{R}^{T\times H}}{\underset{\in}{V}}\right)$$

$$o_k = \mathrm{Att}\left(q_k, K, V\right), \quad k = 1, \dots, \tilde{T}$$

$$\underset{\tilde{T}\times H}{\underset{\in}{O}} = \underset{\text{row\_wise}}{\mathrm{Softmax}}\left(\underbrace{\underset{\tilde{T}\times H}{Q} \cdot \underset{T\times H}{K^T}}_{\tilde{T}\times T} \Big/ \sqrt{H}\right) \cdot \underset{T\times H}{V}$$

Multi Head Attention

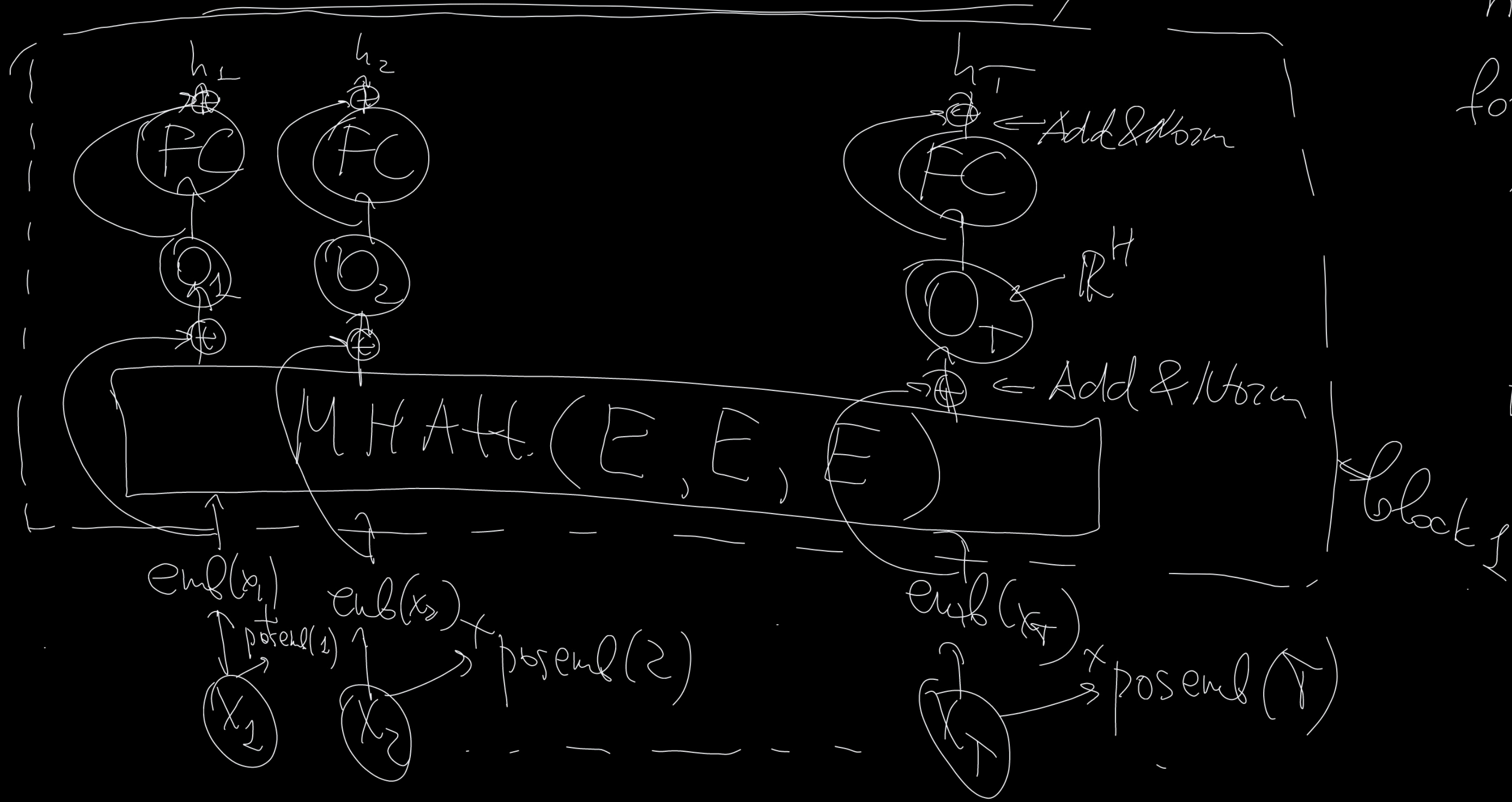
$$\underset{\tilde{T} \times H_{head}}{head_i} = Att(\underset{\tilde{T} \times H}{Q} \underset{H \times H_{head}}{W_i^Q}, \underset{T \times H}{K} \underset{H \times H_{head}}{W_i^K}, \underset{T \times H}{V} \underset{H \times H_{head}}{W_i^V})$$

$\{W_i^Q, W_i^K, W_i^V\}$ – trainable params

$$\underset{\tilde{T} \times H}{O} = \underset{\tilde{T} \times nH_{head}}{MHAtt} = concat(\underset{\tilde{T} \times H_{head}}{head_1}, head_2, \ldots, head_n) \cdot \underset{nH_{head} \times H}{W^O}$$

$\to$ block 2 $\to \dots \to$ block L $\to$ $h_1^L \; \vdots \; h_T^L$

Transformer encoder



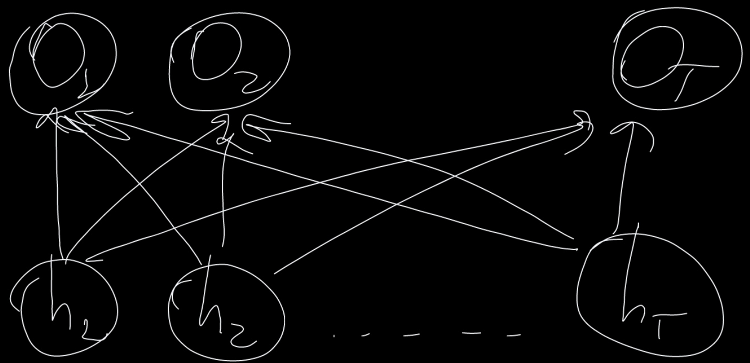
$$h_t^0 = emb(x_t)$$

for $l = 1, 2, \dots, L$:

$$O^l = \mathrm{LayerNorm}(H^{l-1} + \mathrm{MHAtt}(H^{l-1}, H^{l-1}, H^{l-1}))$$

$$H^l = \mathrm{LayerNorm}(O^l + g(O^l W_1 + b_1) W_2 + b_2)$$

$T \times H \quad H \times M \quad M \times 1$



$$h_{t+1} = g(\underset{H\times H}{W^h} h_t + \underset{H\times 1}{W^x} x_t + b)$$

Complexity:

T, H, H_{head}, n

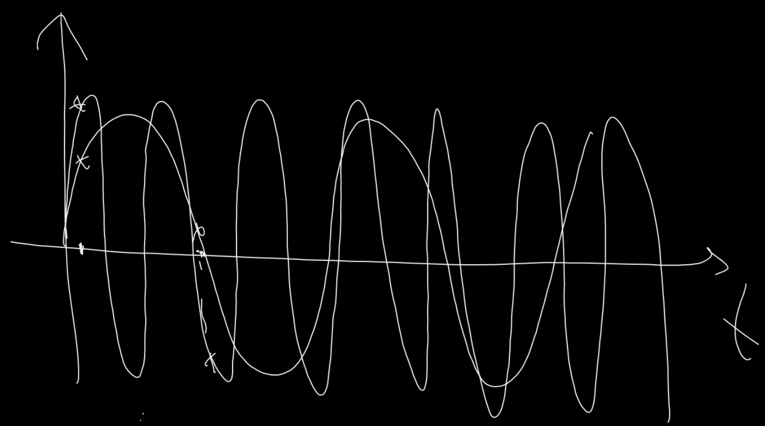
RNN	MH Att.
$O(T\cdot H\cdot H)$	$O(n(T\cdot H\cdot H_{head} + T\cdot T\cdot H_{head}))$
linear w.r.t. T	quadratic w.r.t. T
	$= \{nH_{head} \approx H\} = O(T\cdot H\cdot H + T\cdot T\cdot H)$

Positional Embedding

$$posemb_{t,2i} = \sin(f_i \cdot t) \quad , \; i = 1, \ldots, emb\_dim/2$$

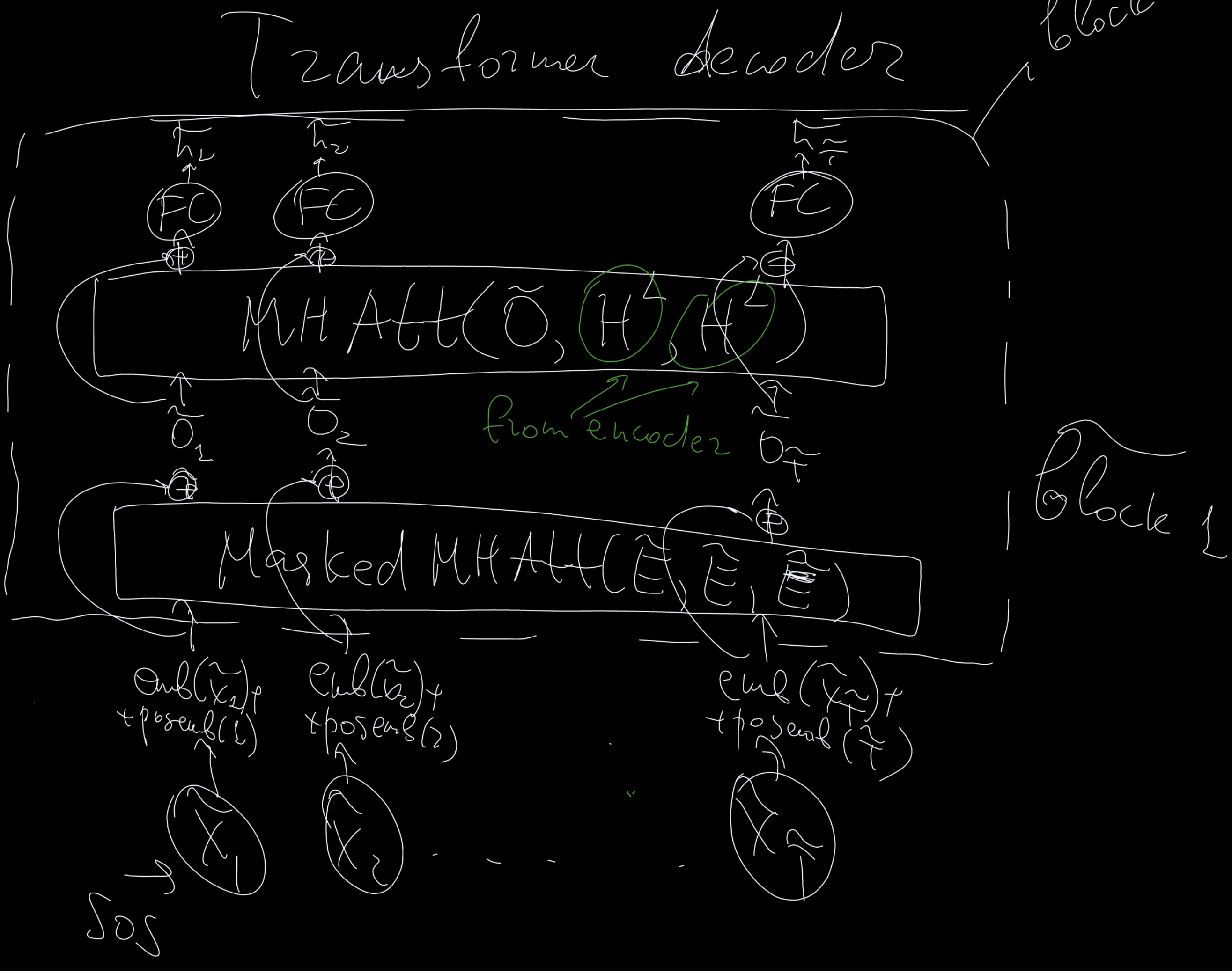
$$posemb_{t,2i+1} = \cos(f_i \cdot t)$$

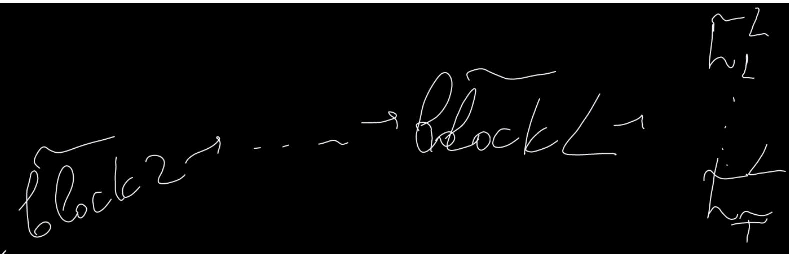
$$f_i = 10000^{\frac{emb\_dim}{2i}}$$



$$posemb_{t+k,2i} = \sin(f_i(t+k)) =$$

$$= \underbrace{\sin(f_i t)}_{posemb_{t,2i}} \cos(f_i k) + \underbrace{\cos(f_i t)}_{posemb_{t,2i+1}} \sin(f_i k)$$





$$\tilde h_t^0 = emb(\tilde x_t)$$

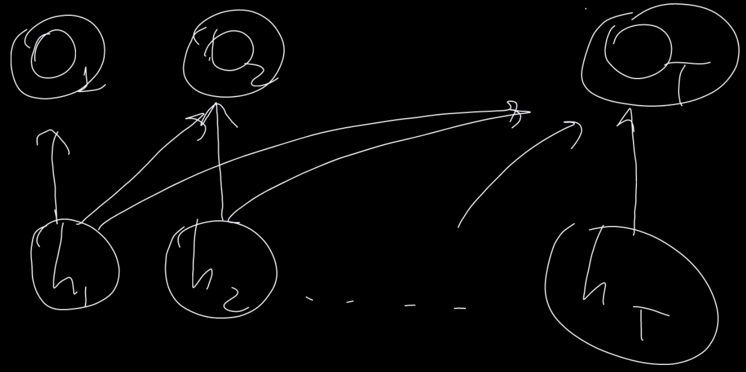
for $l = 1, \dots, L$:

$$\tilde O^l = LayerNorm(\tilde H^{l-1} + MaskedMHAtt(\tilde H^{l-1}, \tilde H^{l-1}, \tilde H^{l-1}))$$

$$\tilde O^l = LayerNorm(\tilde O^l + MHAtt(\tilde O^l, H^L, H^L))$$

$$\tilde H^l = LayerNorm(\tilde O^l + g(\tilde O^l W_1 + b_1) W_2 + b_2)$$

Masked Attention



$$O = \text{Softmax}\left(\frac{H \cdot H^T}{\sqrt{\dim(H)}} + \text{mask}\right) \cdot H$$

$$\text{mask}: \quad \begin{array}{c|cccc} & 1 & \cdots & \cdots & T \\ \hline 1 & 0 & -\infty & \cdots & -\infty \\ \vdots & & 0 & -\infty \cdots & -\infty \\ \vdots & & & \ddots & \vdots \\ T & 0 & & & -\infty \\ & & & & 0 \end{array}$$